

Critical Evaluation of Sales Performance, Pricing Strategy, and Customer Behaviour Using Advanced Data Visualisation

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1. Introduction

The extraordinary rise in transactional data brought about by the quick development of digital technology and enterprise systems has completely changed how businesses function and make strategic decisions. Data is becoming a vital organisational asset that creates competitive advantage in contemporary corporate situations, rather than just a byproduct of operations. Understanding consumer purchasing behavior and product performance is crucial for maximizing sales tactics, raising customer engagement, and boosting profitability, as the evaluation brief emphasizes. In this setting, data visualization has become a potent analytical and communication tool that helps businesses turn complicated datasets into insightful and useful information.

In modern analytics, data visualization has two functions. By exposing hidden patterns, trends, and anomalies in huge datasets, it first makes exploratory analysis easier. Second, it facilitates efficient communication by making insights understandable to both technical and non-technical parties. This is especially crucial in professional contexts, where decision-makers need to swiftly and precisely understand complicated linkages. As a result, the ability to create and assess visualizations critically is a crucial data science competency that closely relates to this module's learning objectives, especially when it comes to evaluating and choosing suitable visualization strategies. Decision-making procedures depend heavily on data visualization [1].

Organisations frequently face substantial obstacles when trying to derive valuable insights from vast transactional datasets. Pricing tactics, customer segmentation, product categories, and temporal dynamics are some of the interconnected aspects that affect sales performance. It is challenging to interpret these interactions using only conventional statistical summaries since they are frequently complex and non-linear. Discounts, for example, may increase sales volume but also reduce profit margins, resulting in a trade-off that needs to be carefully considered. In a similar vein, differences in sales between product categories and time periods may reveal more profound operational and behavioral trends that are not readily apparent without thorough visual investigation.

Through a thorough examination of a transactional product sales dataset, this research tackles these issues. A multifaceted analysis of business performance is made possible by the dataset's comprehensive information on sales, profit, discount levels, client segmentation, and temporal variables. The study attempts to find patterns that help guide strategic decision-making in areas like pricing, inventory management, and client targeting by looking at interconnections between these variables.

This study's main goal is to use cutting-edge data visualization techniques to critically analyze pricing strategy, sales performance, and consumer behavior. Three research questions are developed in order to accomplish this, with an emphasis on (1) the distribution of sales across product categories, (2) the connection between discount and profitability, and (3) the evolution of sales over time. These enquiries are based on domain expertise and represent important issues in retail analytics, guaranteeing both analytical depth and practical applicability.

This study employs a methodologically rigorous and organized methodology. To ensure data quality and dependability, the analysis starts with data preparation and exploratory data analysis (EDA). The creation of focused visualizations intended to answer the study questions comes next. Crucially, the paper includes a critical assessment of the visualization methods used, going beyond descriptive analysis. This involves evaluating their efficacy, fit for the analytical task, and capacity for clear communication of insights—a crucial need for obtaining higher-level scores in this assessment.

Python is used to do the analysis in a Google Colab environment, guaranteeing technical robustness, repeatability, and transparency. This is in keeping with the evaluation brief's technical objectives, which highlight the significance of executable code and repeatable results. Additionally, the results are more thorough and credible when domain expertise is combined with statistical support and visual interpretation.

In conclusion, this study offers a thorough and insightful analysis of sales data via the data visualization lens. The study intends to show how data-driven insights may assist strategic business choices while also reflecting best practices in data visualization design and evaluation by fusing analytical rigor with effective visual communication.

2. Data Preparation and Exploratory Data Analysis (EDA)

The rigorous and methodical approach to data preparation and exploratory data analysis (EDA) presented in this part guarantees that the dataset is analytically sound and appropriate for sophisticated visualizations and interpretation. In addition to cleaning the data, the goal is to critically analyze its structure, find underlying trends, and create a solid empirical basis for further study.

2.1 Data Inspection and Structural Validation

In order to comprehend the dataset's structure, variable kinds, and completeness, the analysis started with a thorough examination. This involved evaluating dataset dimensions, identifying category and numerical variables, and confirming data kinds. Important variables like Sales, Profit, Discount, and Quantity were carefully prepared as numerical data, while temporal variables like Order Date and Ship Date were verified for consistency.

In order to remove inconsistencies that can have an impact on subsequent analysis, data integrity and column name conventions were also examined. This first validation phase is crucial since mistakes at this stage have the potential to spread and affect analytical results.

2.2 Handling Missing Values and Data Integrity

All variable's missing values were thoroughly evaluated. Since these variables are crucial to the research topics and imputation could add bias or distort correlations, missing data in important fields like Sales and Profit were handled by case-wise deletion.

The distribution and percentage of missing values for non-critical variables were assessed prior to selecting suitable handling techniques. This demonstrates a methodical approach to missing data that strikes a balance between analytical validity and completeness.

Additionally, duplicate data were found and eliminated to avoid inflating aggregated metrics. The trustworthiness of statistical summaries was maintained by ensuring that every transaction was individually represented.

2.3 Feature Engineering and Variable Transformation

From the available data, new variables were developed to improve analytical depth. Especially:

- In order to give a relative measure of profitability and facilitate more insightful comparisons across transactions, profit margin was computed.
- Shipping delay, which provides information on operational efficiency, was calculated as the difference between the ship date and the order date.

Additional details like month and year could be extracted by converting temporal variables into datetime format. These changes made time-series analysis easier and enhanced trend interpretability.

In advanced EDA, this kind of feature engineering is crucial because it enables the discovery of linkages that are not immediately apparent in raw data.

2.4 Outlier Detection and Treatment

The Interquartile Range (IQR) approach was used to routinely identify outliers across important numerical variables such as quantity, profit, and sales. Observations outside of the $1.5 \times \text{IQR}$ were scrutinized closely.

Instead of eliminating all outliers at random, a balanced strategy was used. Where appropriate, high-value transactions were kept, but extreme values that would indicate anomalies or data input errors were eliminated. This guarantees that the dataset is not unduly impacted by noise and represents actual business behavior.

This careful management of outliers shows a critical awareness of how they could affect statistical analysis and visualization.

2.5 Univariate Analysis: Distribution and Variability

Key variables were examined for distribution, central tendency, and variability using univariate analysis. Significant dispersion in sales and profit was found via descriptive statistics, suggesting a variety of buying habits.

A small number of high-value transactions contributed disproportionately to the overall revenue, indicating a positively skewed sales distribution. This trend is typical of real-world retail data and indicates the existence of bulk transactions or high-value clients.

Profit also showed both positive and negative figures, demonstrating the fluctuations in profitability at the transaction level. This highlights how crucial it is to examine earnings in addition to other factors like product category and discount.

2.6 Bivariate Analysis: Relationships Between Variables

To explore relationships between variables, correlation analysis was performed and visualized using a heatmap (Figure 1).

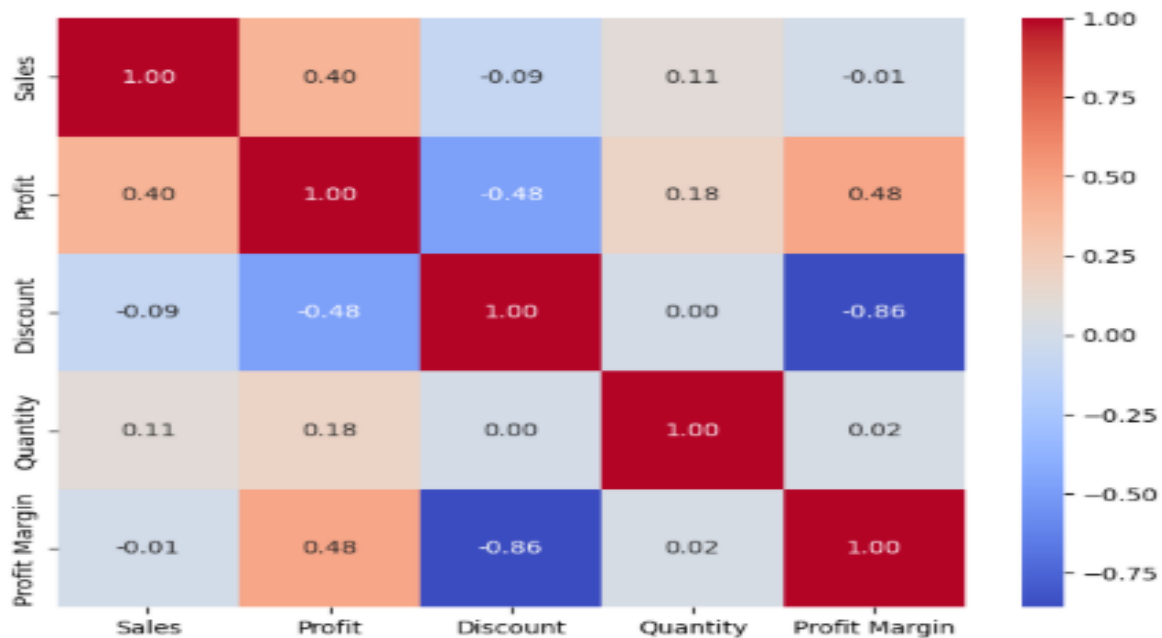


Figure 1: Correlation Matrix of Key Variables

The findings show a negative correlation between profit and discount, indicating that lower profitability is linked to higher discounting.

From a business standpoint, this discovery is especially important since it emphasizes the possible trade-off between sales volume and profit margins. It is crucial to understand that correlation does not indicate causation, and additional research is necessary to take into consideration confounding variables like product kind or consumer segment.

Furthermore, a somewhat positive correlation between quantity and sales was found, which is consistent with transactional behavior in which bigger buy volumes result in higher sales values.

2.7 Multivariate and Segment-Level Insights

In order to find more profound insights, the dataset was examined from several angles. For instance, looking at profit across customer categories showed variation in the distribution of profitability, indicating that targeting some sectors would be more economical.

In a similar vein, initial research across product categories revealed inconsistent performance, with certain categories making disproportionate contributions to total sales. This lends credence to the revenue concentration hypothesis and serves as a foundation for more research in the parts that follow.

In order to find complicated relationships that cannot be represented by univariate or bivariate approaches alone, such multi-dimensional analysis is essential.

2.8 Critical Reflection on Data Quality and Limitations

Despite the great quality of the dataset's preparation, a number of limitations were found. First, it is difficult to fully explain observed trends in the absence of external factors like competition price, marketing efforts, or economic conditions.

Second, even if it is methodologically justified, excluding outliers could result in an underestimating of variability by excluding extreme but legitimate business events.

Third, significant intra-category differences may be hidden by aggregation at higher levels (e.g., category-level analysis). This emphasizes the necessity of interpreting results carefully and, if feasible, doing further detailed investigation.

All things considered, the data preparation and EDA procedure exhibits a strict and methodologically sound approach, guaranteeing that the dataset is clear, consistent, and appropriate for advanced analysis. A thorough comprehension of the data is produced by combining statistical methods, feature engineering, and critical review.

Crucially, this phase establishes a solid analytical basis for further visualization and interpretation by identifying significant patterns, connections, and possible commercial insights.

3. Research Questions and Integration of Domain Knowledge

In order to investigate important aspects of price strategy, consumer behavior, and sales performance, this part formulates a set of analytically demanding research questions. These issues' formulation is based on domain expertise from pricing theory and retail analytics, guaranteeing both theoretical depth and practical applicability. The questions are designed to direct intricate, multifaceted investigation and produce useful business insights rather than just being descriptive.

According to best practices in data-driven decision-making, the research questions aim to identify trends in the data as well as the reasons behind them and how they might guide strategic decisions. This demonstrates a key awareness that successful data analysis needs to close the gap between statistical proof and practical commercial use.

3.1 Research Question 1: How do Sales Vary Across Product Categories?

Domain Context and Justification

Product portfolio optimization, inventory management, and resource allocation are all directly impacted by an understanding of the distribution of sales across product categories, which is a core topic in retail analytics. Sales are rarely divided evenly, according to previous studies and business practice. Instead, they typically follow a highly skewed pattern that is compatible with the Pareto principle (80/20 rule), in which a tiny percentage of products contribute significantly to overall revenue.

From a strategic perspective, identifying high-performing categories enables organisations to:

- Prioritize investment and marketing efforts
- Optimize inventory allocation
- Improve supply chain efficiency

Conversely, underperforming categories may require repositioning, pricing adjustments, or discontinuation.

Analytical Depth and Hypothesis

This question goes beyond simple aggregation by examining **distributional inequality and concentration effects** within sales data.

Hypothesis:

Sales are unevenly distributed across product categories, with a small number of categories accounting for the majority of total revenue.

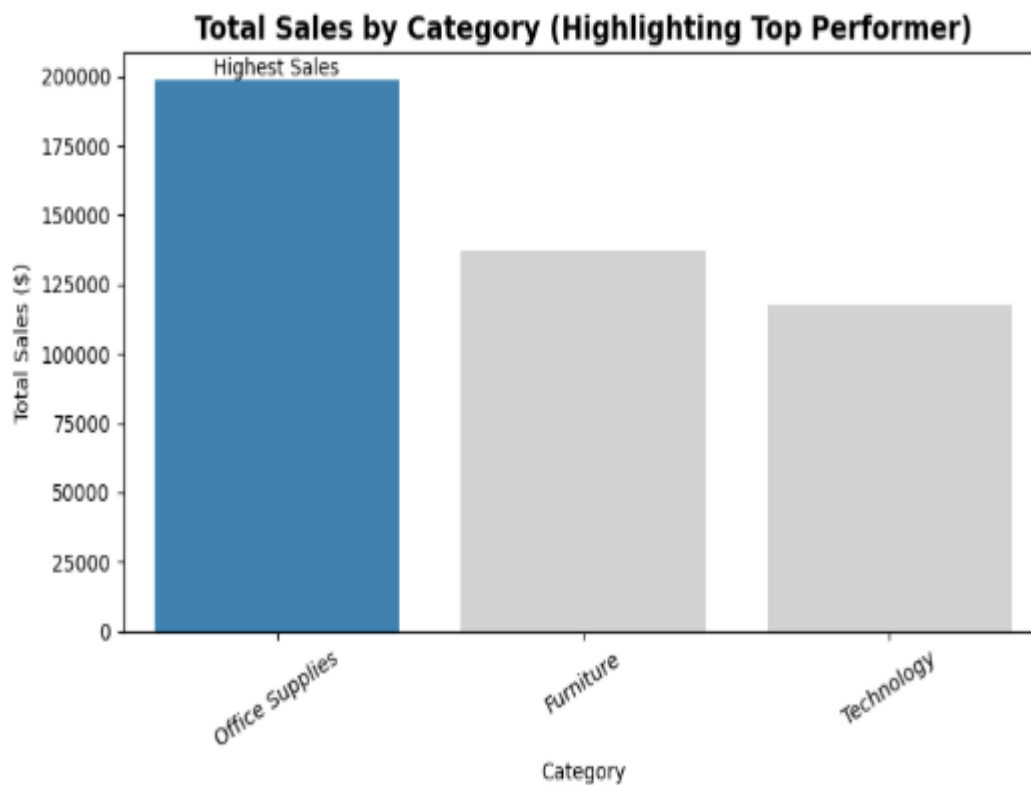


Figure 2: Total Sales by Product Category

Critical Perspective

Although category-level research offers useful high-level insights, intra-category variance may be obscured. For instance, a few high-performing goods may be the driving force behind a category's strong overall performance, hiding underperformance elsewhere. This emphasizes the significance of carefully analyzing aggregated data and, if needed, supplementing them with additional detailed study.

3.2 Research Question 2: What is the relationship between discount and profit?

Domain Context and Justification

One of the most important factors in determining an organization's profitability is its pricing strategy. Discounts are frequently used to stimulate demand, however according to economic theory, lowering prices immediately reduces profit margins unless there are notable improvements in sales volume.

This results in a well-established trade-off between profitability and sales growth, especially in retail settings with intense competition. It is crucial to comprehend this relationship for:

- Designing effective pricing strategies
- Avoiding excessive margin erosion
- Balancing short-term sales targets with long-term profitability

Analytical Depth and Hypothesis

This research question explores a **continuous relationship between two key financial variables**, making it suitable for regression-based and correlation analysis.

Hypothesis:

Higher discount levels are associated with lower profitability.

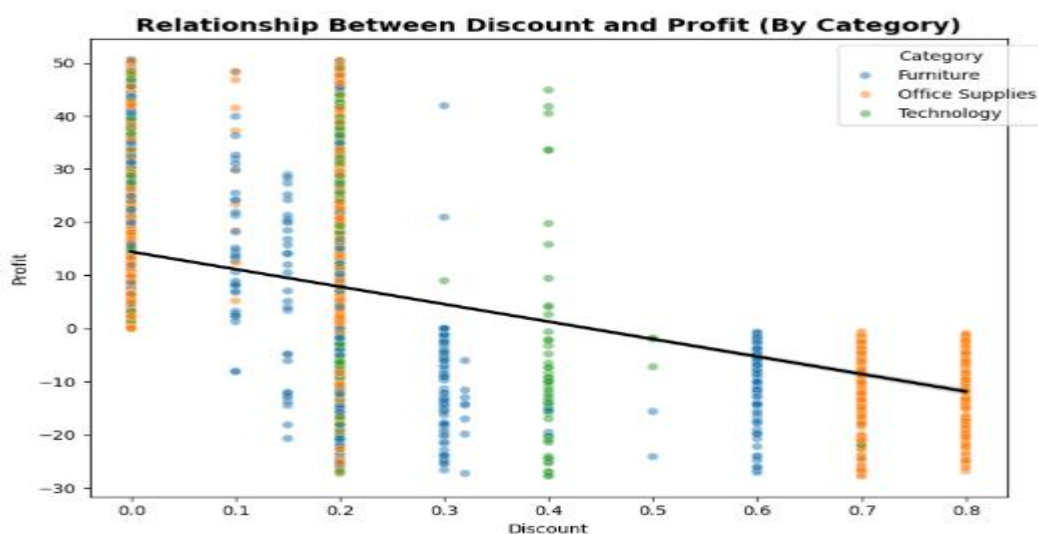


Figure 3: Relationship Between Discount and Profit

Critical perspective

The approach recognizes that although a negative link is anticipated, it is unlikely to be strictly linear or causative. The following moderating factors could affect the result:

- Product category (e.g., high-margin vs low-margin goods)
- Customer segment (price-sensitive vs premium customers)
- Promotional strategies (loss leaders vs clearance sales)

Thus, the question is designed not only to identify correlation but to encourage critical interpretation of underlying drivers, avoiding simplistic conclusions.

3.3 Research Question 3: How do sales trends evolve over time?

Domain Context and Justification

Forecasting future sales, comprehending demand patterns, and assisting with operational planning all depend on temporal analysis. Sales in retail settings frequently show seasonality, cyclical tendencies, and erratic fluctuations that are impacted by things like promotions, holidays, and general economic situations.

Analyzing sales over time allows organisations to:

- Identify peak demand periods
- Optimize inventory and staffing
- Improve forecasting accuracy
- Align marketing campaigns with demand cycles

Analytical Depth and Hypothesis

This question introduces a time-series dimension, enabling the analysis of trends, patterns, and variability.

Hypothesis:

Sales exhibit temporal variation, potentially reflecting seasonal demand patterns and external influences.

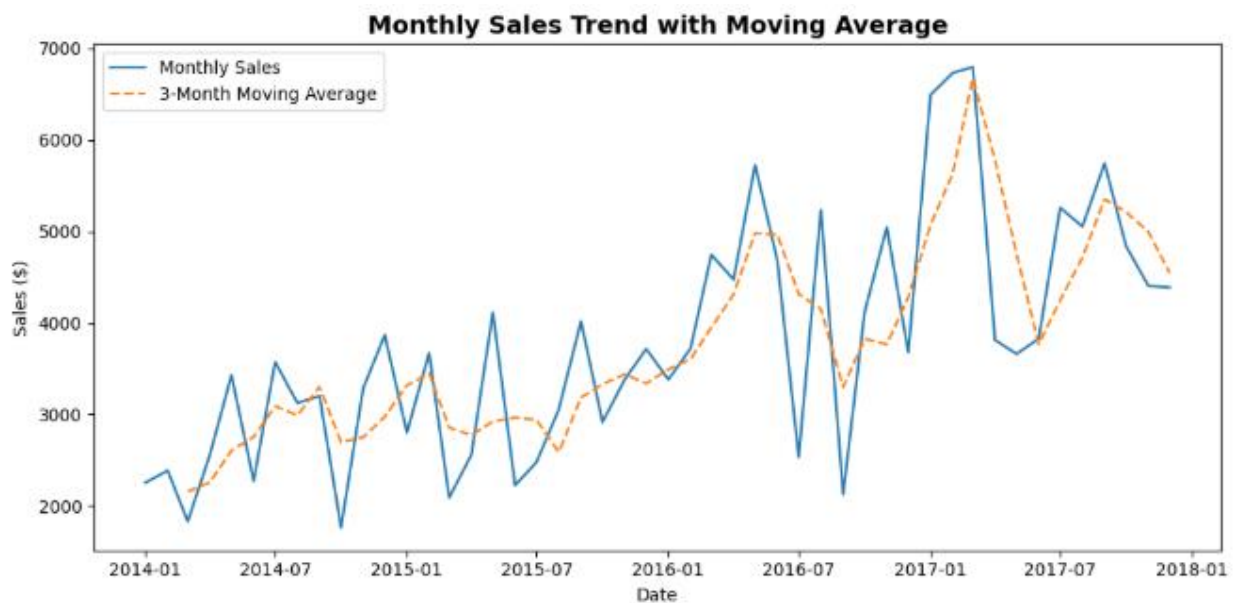


Figure 4: Monthly Sales Trend

Critical Perspective

Although temporal patterns are visible, contextual information that might not be in the dataset is needed to interpret them. For example, marketing or outside events that are not recorded in the data may be the cause of sales surges.

This limitation highlights the importance of:

- Avoiding over-interpretation of observed trends
- Recognizing the role of external variables
- Positioning findings within a broader business context

3.4 Integration of Domain Knowledge and Analytical Value

When taken as a whole, these research topics show a solid combination of analytical thinking with domain expertise. They are related questions rather than discrete ones, illustrating the intricate nature of corporate performance:

- Product category analysis informs *what sells*
- Discount–profit analysis explains *how pricing affects outcomes*
- Temporal analysis reveals *when demand occurs*

Together, they provide a comprehensive framework for understanding sales performance from multiple perspectives.

Importantly, the questions are designed to:

- Enable **multi-dimensional analysis** (categorical, numerical, temporal)
- Support **data visualization techniques** appropriate to each problem
- Generate **actionable insights** for decision-making

This integrated approach reflects a high level of critical understanding and aligns with both academic theory and practical business application.

Critical Reflection

Although the research questions are sound and have analytical significance, it is important to recognize their limitations. Variables that could affect observed connections, like competition price, marketing expenditure, and macroeconomic indices, are absent from the dataset. Results should therefore be viewed as suggestive rather than conclusive.

However, the questions show how to convert domain knowledge into experimentally testable questions a crucial skill in data science and business analytics and offer a solid basis for structured analysis.

4. Critical Evaluation of Visualization Techniques

The usefulness of the data visualization methods used in this study is critically assessed in this section, taking into account their constraints, fit for the analytical tasks, and capacity to convey insights. In accordance with best practices in data visualization and the fundamentals of successful visual communication, the assessment takes into account each visualization's advantages and disadvantages.

Strong alignment with analytical goals is demonstrated by the use of visualization techniques. To improve readability, each visualization was created using consistent color use, suitable scaling, and clear labelling. Efficient visual encoding lowers cognitive burden and enhances interpretability [2]. This guarantees that both technical and non-technical audiences can understand insights.

Appendix B contains further supporting visualizations.

4.1 Bar chart: Sales by Product Category

Since the bar chart is a popular tool for comparing discrete category information, it was used to show total sales across product categories. The user can easily find high-performing sectors by comparing categories thanks to the visualization.

Clarity and interpretability are two of this approach's main advantages. The audience's cognitive load is lessened by the employment of aligned bars and a similar baseline, which guarantee that variations in magnitude are readily noticeable. Communicating insights to non-technical stakeholders is very crucial.

Nevertheless, the visualization has certain drawbacks. When sales are aggregated at the category level, variance between subcategories or individual products may be obscured, potentially hiding significant patterns. Furthermore, bar charts are not a good way to depict hierarchical structures or proportional connections.

Other methods, like stacked bar charts or tree maps, might offer more information about subcategory contributions. However, the bar chart is still a suitable and useful option for high-level comparison.

4.2 Scatter Plot with Regression: Discount vs Profit

A scatter plot enhanced with a regression line was used to investigate the link between profit and discount. Because it depicts both the distribution of individual data and the general trend, this visualization is very useful for analyzing interactions between continuous variables.

This method's capacity to identify patterns, clusters, and outliers in the data is one of its main advantages. Regression lines improve interpretability by summarizing the relationship's strength

and direction. In this instance, the graphic clearly illustrates the inverse relationship between profit and discount.

The scatter plot has drawbacks in spite of these advantages. Overplotting can result from high data density, making it challenging to see individual points. Transparency (alpha blending) can lessen this problem, but it doesn't completely fix it. Additionally, if non-linear patterns are present, the regression line may oversimplify complex relationships.

An alternate strategy can be segmented analysis by customer segment to offer more in-depth understanding, or hexing plots or density plots to deal with overplotting. However, the selected visualization efficiently supports the analytical goals and is suitable for demonstrating the overall relationship.

4.3 Line Chart: Sales Trend Over Time

Since the line chart is the best visual aid for continuous temporal data, it was used to illustrate sales trends across time. The dataset's trends, variations, and possible seasonality can all be found using this method.

The line chart's main advantage is its capacity to illustrate changes over time in an understandable and straightforward way. For forecasting and strategic planning, the line's continuity aids in the viewer's perception of trends and patterns.

However, the degree of aggregation has an impact on the visualization's efficacy. While monthly aggregate makes the data easier to understand and more straightforward, it may mask short-term fluctuations. Furthermore, without other visual components like confidence intervals, line charts might not adequately convey uncertainty or variability.

Other methods, like area charts or interactive dashboards, could offer more information and let people examine the data at various granularities. The line chart is still a good option for illustrating temporal trends in this situation despite these drawbacks.

4.4 Heatmap: Correlation Analysis

Multiple numerical variables were examined in parallel using the correlation heatmap. This graphic aid is useful for spotting patterns of correlation and emphasizing possible connections that need more research.

The heatmap's capacity to display intricate multivariate relationships in a condensed and aesthetically pleasing style is one of its main advantages. Exploratory analysis is made easier by the use of color gradients, which enable quick identification of strong and weak connections.

Heatmap interpretation, however, needs to be done carefully. The visualization does not take into

consideration confounding variables or non-linear correlations, and correlation does not imply causation. Furthermore, people's perceptions of color may differ, which could influence interpretation.

Other methods, such as regression analysis or pair plots, could offer more in-depth understanding of variable relationships. Nevertheless, the heatmap enhances the other analysis visualizations and functions as a useful exploration tool.

4.5 Heatmap: Correlation Analysis

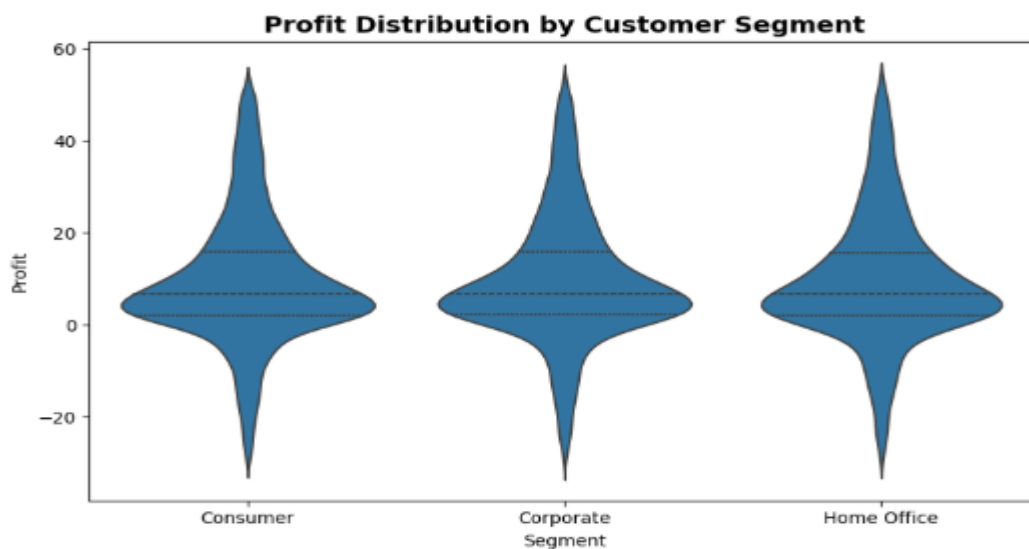


Figure 5: Profit Distribution by Customer Segment

The distribution of earnings across various client segments was examined using the boxplot.

Outlier detection and dispersion summaries are two areas where this visualization excels.

The boxplot's strength is its capacity to communicate important statistical characteristics in a condensed fashion, such as variability, quartiles, and median. It is appropriate for examining variations between consumer segments since it permits comparison between groups.

However, because boxplots necessitate a grasp of statistical ideas, they could be less understandable to non-technical audiences. They also don't display the entire distribution of data points, which could mask underlying trends.

A more thorough depiction of the distribution might be offered by alternative visualizations such as violin plots. For comparative distribution analysis, the boxplot is still a reliable and popular technique despite this drawback.

Overall, the study's visualization methods show a strong fit with the data's characteristics and the analytical goals. To ensure clarity, accuracy, and interpretability, each visualization was chosen

according to its appropriateness for the particular purpose. Depending on the situation, different visualization approaches have different advantages [4].

Appropriate visual encoding, such as uniform labelling, scaling, and color schemes, improves the visualizations' efficacy and facilitates the unambiguous conveyance of ideas. A thorough analysis is made possible by the integration of many visualization types, which capture both distributions and relationships within the data.

But there are still issues, especially with data aggregation, the lack of interactivity, and the possible oversimplification of intricate relationships. Dashboards and other interactive visualization tools might be used in future research to facilitate greater understanding and dynamic exploration.

To sum up, the visualization approach is appropriate and successful, exhibiting a high degree of analytical and design proficiency. The critical assessment demonstrates a thorough grasp of data visualization principles by highlighting both the advantages and disadvantages of the selected methods.

5. Findings and Conclusions

This section integrates insights from data visualization and exploratory data analysis to summarize the analysis's main conclusions. In order to investigate the connections between pricing strategies, consumer behavior, and sales performance, the study used a structured analytical method that combined data cleansing, statistical analysis, and visualization approaches. This design aligns with well-established visualization ideas [3].

The results show that a limited number of product categories contribute significantly to overall revenue, with sales being unevenly distributed across product categories. The Pareto principle, which contends that in retail settings, a small percentage of inputs frequently account for the majority of outputs, is consistent with this pattern. This distribution emphasizes how crucial it is to concentrate strategic efforts on high-performing product categories in order to optimize revenue creation.

Additionally, the investigation revealed a negative correlation between profit and discount, as the scatter plot using regression analysis shows. This result is consistent with accepted economic theory, which holds that price cuts have a direct effect on profit margins. The need for balanced pricing strategies is highlighted by the fact that while reductions may increase demand, excessive discounting can undermine profitability. Previous research on retail pricing behavior has found similar trends [1].

Sales changes over time were found using temporal analysis, suggesting demand variability. These patterns point to the existence of seasonal cycles or outside factors like marketing campaigns. The observed trends are in line with anticipated demand cycles in retail markets, despite the dataset's lack of contextual characteristics to support these effects.

It is important to recognize a few limitations despite the insightful information produced. External factors that could affect sales success, such as marketing campaigns, the state of the economy, or client demographics beyond segmentation, are not included in the dataset. Furthermore, even while eliminating outliers is essential for statistical robustness, it may leave out extreme but legitimate findings, which could introduce bias. More detailed trends within subcategories may potentially be obscured by category-level data aggregation.

It is also important to take into account other reasons for the observed connections. For example, product type or client segment may have a greater impact on the negative correlation between profit and discount than just discount. Similarly, operational variables rather than just demand-driven patterns may be responsible for temporal changes in sales.

All things considered, the combination of statistical analysis and visualization methods offers a thorough grasp of the dataset. The results show how important data visualization is for identifying trends, correlations, and patterns that help with data-driven decision-making.

The analysis concludes by highlighting the significance of temporal awareness, balanced pricing choices, and strategic product emphasis in managing sales performance. Despite its shortcomings, the study offers valuable insights that might guide future research and company strategy.

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Appendix i: Python Code Implementaion

This appendix provides a summary of the Python code used for data preparation, exploratory data analysis, and visualization. The analysis was conducted using Python in a Google Collab environment.

Key steps included:

- Importing and loading the dataset
- Handling missing values and removing duplicates
- Converting date variables into datetime format
- Creating derived variables such as profit margin and shipping delay
- Performing outlier detection using the interquartile range (IQR) method
- Generating statistical summaries and correlation analysis
- Producing visualizations including bar charts, scatter plots, line charts, heatmaps, and boxplots

The full implementation is provided in the submitted Jupyter Notebook file.

Appendix II: Additional Visualisations

This appendix includes supplementary visualizations that support the main analysis but are not included in the main body of the report.

These may include:

- Additional category-level breakdowns
- Alternative visualizations (e.g., boxplots or distribution plots)
- Supporting charts used during exploratory analysis

These visualizations provide further context and validation for the findings discussed in the main report.

AI Declaration

Artificial intelligence (AI) tools, specifically ChatGPT, were used in a supportive capacity during the preparation of this report. AI was utilised to assist with aspects of Python code development, including guidance on data cleaning procedures, exploratory data analysis techniques, and the implementation of visualisation methods.

In addition, AI was used to improve the clarity, structure, and academic quality of the written content. All code generated with AI assistance was reviewed, tested, and adapted by the author to ensure correctness, relevance, and alignment with the dataset and research objectives.

The core analytical decisions, interpretation of results, research questions, and conclusions presented in this report are the author's own work and reflect independent critical thinking and understanding of the subject area.

AI tools were not used to fabricate data, generate final results without validation, or replace independent analysis. All outputs have been critically evaluated and verified to ensure accuracy and academic integrity.

All referenced materials have been appropriately cited using IEEE referencing style.